

**Exponent Laws with Variable Bases**Question Bank 3.T1 **ANSWER KEY**

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Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression as a single monomial. Assume no variable equals 0.

1.  $x^4 \cdot x^9$   
 $x^{13}$

2.  $(n^5)^6$   
 $n^{30}$

3.  $(3y^4)^3$   
 $27y^{12}$

The 3 is cubed as well.

4.  $\frac{a^9}{a^3}$   
 $a^6$

5.  $(-6x^2)(2x^7)$   
 $-12x^9$

6.  $\frac{20p^5q^3}{5p^2q}$   
 $4p^3q^2$

Coefficients divide; each variable's exponents subtract.

For questions 7 and 8, apply more than one law. Write each answer as a single monomial.

7. Write an equivalent expression.

a.  $(2x^3y)(5x^2y^4)$

$10x^5y^5$

b.  $(4m^2n^3)^2$

$16m^4n^6$

c.  $(a^2b)^3 \cdot ab^4$

$a^7b^7$

$a^6b^3 \cdot ab^4$ .

8. Write an equivalent expression.

a.  $\frac{(3m^2n)^3}{9m^4n}$

$3m^2n^2$

$27m^6n^3$  over  $9m^4n$ .

b.  $\frac{x^8y^5}{x^8y^2}$

$y^3$

$x^0 = 1$ .

c.  $\frac{(2k^4)^3}{4k^7}$

$2k^5$

$8k^{12}$  over  $4k^7$ .

**Working backwards.**

9. Find the value of  $n$  that makes each statement true.

a.  $x^3 \cdot x^n = x^{11}$

$n = 8$

b.  $(y^n)^4 = y^{20}$

$n = 5$

c.  $\frac{m^9}{m^n} = m^2$

$n = 7$

$9 - n = 2$ .

10. A rectangle has length  $6x^4$  and width  $2x^3$  .

a. Write its area as a single monomial.

$$12x^7$$

*length  $\times$  width; coefficients multiply, exponents add.*

b. A square has side length  $4y^5$  . Write its area as a single monomial.

$$16y^{10}$$

*$(4y^5)^2$  — the 4 is squared too.*

11. Which expression is equivalent to  $(5m^3)^2$  ?

☐ A.  $5m^6$

☐ B.  $25m^5$

☐ C.  $10m^6$

☒ D.  $25m^6$

**D**

12. Select ALL of the expressions that are equivalent to  $x^{12}$  .

☒ A.  $(x^4)^3$

☐ B.  $x^2 \cdot x^6$

☒ C.  $(x^6)^2$

☒ D.  $\frac{x^{15}}{x^3}$

☐ E.  $x^4 \cdot x^3$

☒ F.  $x^4 \cdot x^8$

**A, C, D and F**